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Optical clock networks

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Within the last decade, optical atomic clocks have surpassed the best cesium clocks, which are used to realize the unit of time and frequency, in terms of accuracy and stability by about two orders of magnitude. When remote optical atomic clocks are connected by links without degradation in the clock signals, an optical clock network is formed, with distinct advantages for the dissemination of time, geodesy, astronomy and basic and applied research. Different approaches for time and frequency transfer in the microwave and optical regime, via satellites and free-space links, optical fibre links, or transportable optical atomic clocks, can be used to form a hybrid clock network that may allow a future redefinition of the unit of time based on an optical reference transition.

Atomic clocks are among the very best measuring devices and are indispensable in everyday life for accurate timekeeping in the worldwide system of coordinated universal time (UTC; http://www.bipm.org/en/bipm-services/timescales), for global navigation satellite systems[1] and for the synchronization of grids for electrical power or cellular phones[2]. They allow testing of basic theories such as general relativity or quantum electrodynamics and their foundations, for example, Lorentz symmetry[3], the Einstein equivalence principle[4] or the constancy of fundamental constants[5]. Furthermore, atomic clocks are finding numerous research applications in geodesy[6] and astronomy, for example[7,8]. Today, the best optical atomic clocks[9,10] (Fig. 1 and Box 1) reach a fractional systematic uncertainty at the low 10^{-18} level[11–13] and fractional instabilities in the same regime below 10^{-17} (refs 14,15) within reasonable averaging times, thereby outperforming cesium atomic clocks that are used to define the second in the International System of Units (SI) by two and three orders of magnitude, respectively. These achievements open new fields of research[16–19] and applications[20] and could stimulate a new definition of the SI second[21] and more accurate and stable timescales. It has been shown that today's optical atomic clocks could lead to a better local approximation of UTC[22,23]. The progress in improved timescales will critically depend on how the frequencies of remote optical atomic clocks are compared. I refer to the linked optical atomic clocks as an optical clock network (OCN) and will describe the different ways (Fig. 2) to establish such networks. In this Review, I will discuss these networks, the ways they are connected, their applications as well as future opportunities.

Time and frequency transfer via satellites by microwave signals provides today's established systems to compare distant atomic clocks (Box 2). Two standard techniques are employed. The first makes use of the reception of microwave signals (in the L-band) with time information from global navigation satellite systems, such as the US Global Positioning System (GPS), the Russian GLONASS, the European Galileo, or the Chinese Beidou, to name only those that have nearly global coverage. The method utilizes the time signals received from the satellites at the same time in different laboratories (common view), corrected for the path delay. These signals are compared in the distant laboratories with the respective local clocks. Exchanging the measured time differences allows the difference between the two distant clocks to be derived. By evaluating the carrier phase of the GPS signal an instability of 10^{-15} (Box 3) can be reached within an integration time of 1 day[24] or less[25] (Fig. 3). A second more accurate means of comparing clocks is by using the two-way satellite time and frequency transfer (TWSTFT)[26], where two stations with clocks at different locations exchange their signals via a satellite on a geostationary orbit simultaneously and along the same signal path. The statistical uncertainty of a frequency comparison over transcontinental distances has been shown[27] to be limited to about 10^{-15} in most cases even though much lower uncertainties could be reached in principle. With increased bandwidth, a campaign in the framework of a European project[28] has been trying to reduce this level. A two-way carrier-phase-based data analysis provided a measurement precision of 2×10^{-13} at 1 s and the long-term stability reached the 10^{-16} level[29], with the prospect of reaching the 10^{-17} level[30]. It is planned within the Atomic Clock Ensemble in Space (ACES) project[31] to take a cold atomic cesium clock and a hydrogen maser to the International Space Station (ISS) to allow for time and frequency transfer using several microwave terminals in timing laboratories in France, Germany, England, the US, Japan and elsewhere. The aim is to reduce the relative frequency instability below 10^{-17} for several days of averaging. The use of larger bandwidth or software-based equipment for processing standard TWSTFT signals may also improve the relative frequency instability[32,33].

Optical free-space transfer of frequency and time would allow such instabilities to be reached much faster than with microwave links. In the Time Transfer by Laser Link (T2L2)[34] experiments, asynchronous pulsed laser beams from ground stations are retro-reflected from low-Earth-orbiting satellites, for example Jason II, and the start and return time of each laser pulse is recorded. The arrival time of each pulse is detected on the satellite and time tagged with an onboard clock. Links between distant clocks with time instabilities of a few picoseconds and an uncertainty for time transfer lower than 140 ps with a coverage factor $k = 2$ (refs 34,35) have been operated. There are also plans to use the so-called European Laser Timing instrument[36] together with the T2L2 ground stations as part of the ACES experiment[37] where the arrival time of the signal is detected and compared with the clock onboard the ISS. This link will require several days of integration time to reach a fractional instability of 10^{-17}.

From experiments on the ground it has been demonstrated that a future coherent optical link from ground to satellite through the turbulent atmosphere seems feasible[38]. Theoretical modelling of the influence of turbulence indicated that a frequency stability better than 2×10^{-17} in a 1 s integration time is possible and has good prospects of reaching the sub-10^{-18} level after a few minutes[39]. A recent two-way experiment on the ground has shown that two optical

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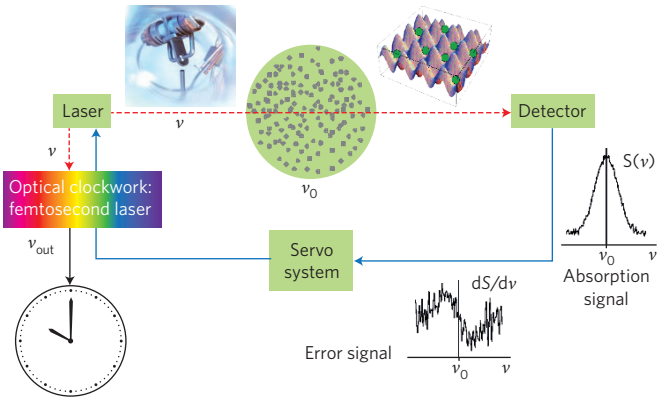


Figure 1 | Schematic of an optical atomic clock where optical transitions in atoms or ions are used to stabilize the frequency of a narrow-band laser to the centre of the transition. The servo system keeps the laser frequency ν as close as possible to the centre frequency ν_0 of the absorption line. The insets show an ion trap (left) and an optical lattice (right) to store a single ion or a large number of neutral atoms, respectively. ν is divided in a frequency comb to derive radiofrequencies ν_{out} or second pulses (see Box 1). S represents the detected signal generated via light absorption.

timescales can be synchronized across a free-space link including atmospheric turbulence within the picosecond range[40] and can fall below 10^{-18} of fractional instability at 1,000 s (Fig. 3).

Optical fibre links have been utilized for time and frequency transfer on various occasions[41–48]. For an OCN, however, that aims to compare and link the optical atomic clocks with the highest accuracy, the existing network has to be modified. First, the optical path length of the fibre has to be stabilized, which can be done by sending the frequency back on a reciprocal path between the two stations, thus measuring and correcting any occurring frequency shift[49]. In dedicated fibre links over distances where the attenuation in the fibre for a given signal-to-noise level is low enough, the phase of the received signal could be tracked unambiguously[50–53]. For longer distances, the transferred frequency has to be phase-coherently amplified. This requires extensive technical manipulation of the existing fibre infrastructure. Here, two different approaches are used, either a 'dark fibre' or a 'dark channel'. The first approach relies on a fibre that is solely used for the frequency transfer. In the second approach, wavelength-division multiplexing is employed where a number of optical carrier signals travel in a single optical fibre in different channels and only one channel is free from Internet traffic. The necessary phase-coherent two-way amplification of the carrier frequency is accomplished by a bypass[54] with or without amplifier[47] at suitable distances along the fibre link. The dark fibre approach is more expensive but has the advantage that more extensive modifications of the fibre lines are possible, for example, Brillouin amplification[48] with higher power, thereby reducing the number of repeater stations considerably. Both technologies were used on the transnational link[55] that has been used to compare the Sr optical lattice clocks of LNE-SYRTE in Paris and Physikalisch-Technische Bundesanstalt (PTB) in Braunschweig. The two optical lattice clocks agreed within their combined fractional uncertainties of 5×10^{-17} (Box 3). This link has been extended recently from France to the National Physical Laboratory in Teddington, UK[21] and will be used to compare the optical atomic clocks of the National Physical Laboratory, LNE-SYRTE and PTB. Analysis of essential parts of such fibre links has shown that they can support frequency comparisons with a fractional uncertainty down to the 10^{-20} regime[48] or below[56], and therefore will not limit the accuracy of optical atomic clock comparisons.

Transportable optical atomic clocks are being developed in several places. A transportable Sr lattice clock to be used for comparing remote optical atomic clocks and for novel applications, for example, in geodesy, has been developed[57] that is operated in a transportable walk-in container that provides the same stable environmental

Box 1 | Optical atomic clock.

In an optical atomic clock (Fig. 1), the frequency of a laser is closely tuned to a very narrow transition line in an atomic, ionic or molecular absorber. The excitation probability of the absorber as function of the frequency of the laser exhibits a narrow feature that is measured most commonly by observing the fluorescence emitted that directly indicates the change of the atomic state. From the detected absorption signal an error signal is generated, for example, by differentiation. The error signal is used by a mostly digital servo loop to steer the frequency of the laser to coincide with the centre of the detected signal. The optical frequency of the stabilized laser can be used as a frequency standard. A femtosecond frequency comb allows one to generate any user-defined optical frequency or a radiofrequency, which, in turn, is counted to produce the pulse-per-second of the clock.

To achieve a long interrogation time and the associated small linewidth of the absorption line, the absorbers are kept against gravity in an electrodynamic trap (ions) or in an optical lattice (neutral absorbers). In the first case, only one or at most a few ions can be kept in the field-free centre region of the trap, whereas in the second case, a few hundred to up to 10^4 neutral absorbers can be stored, which leads in general to an improved signal-to-noise ratio.

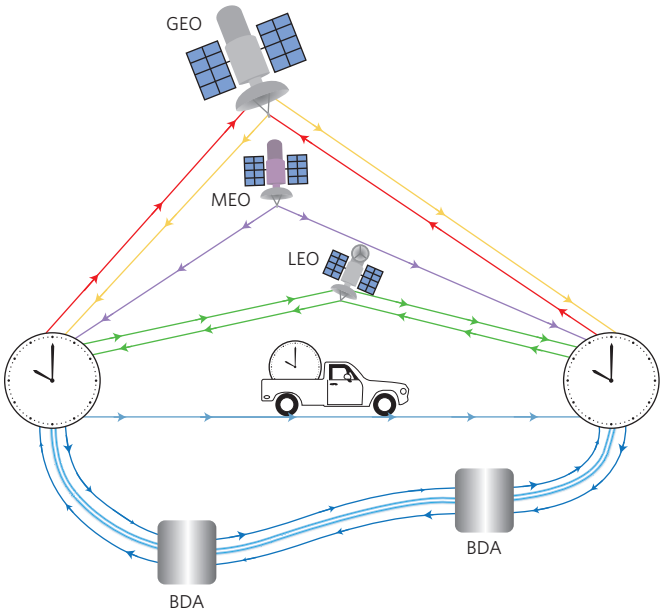


Figure 2 | Different methods can be used to compare the frequencies of optical atomic clocks. Two-way satellite time and frequency transfer to satellites on a geostationary orbit (GEO) with different uplink and downlink frequencies in the microwave range (red and yellow arrows) is widely used for time and frequency comparisons. The common-view technology uses clocks on board global navigation satellite systems on a medium Earth orbit (MEO; purple arrows). Optical links to satellites on a low Earth orbit (LEO; green arrows) such as the International Space Station are under development. Fibre links (dark blue arrows) with bi-directional amplifiers (BDA) represent point-to-point connections and are a mature technology. Transportation of optical atomic clocks (light blue arrows) with lowest uncertainties is still in the pilot phase.

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conditions at each operation site. The total systematic uncertainty is estimated to be 7×10^{-17} and the frequency instability is found to be 1.3×10^{-15} $(\tau\ \mathrm{s}^{-1})^{-1/2}$, where τ is the integration time. A group supported by the European Union Seventh Framework programme is developing a transportable standard[58] based on a Sr lattice clock that could also be used for space applications[59]. A transportable clock in a volume of about 0.1 m^3 based on a trapped single Ca^+ ion with the physics package with low systematic uncertainty of 7.7×10^{-17} and a fractional instability of 2.3×10^{-14} $(\tau\ \mathrm{s}^{-1})^{-1/2}$ has also been reported recently[60].

Applications of OCNs and future prospects

All the necessary ingredients for local, national and even transnational OCNs have been developed and are used for various purposes. Particular OCNs have been set up for example in Japan[61,62], Europe (Fig. 4) and the US[63,64] for highly accurate frequency comparisons and other purposes as discussed below.

Timescales could be improved with respect to stability and availability as a direct consequence of the existing and future OCNs when more stable optical atomic clocks will be included in the generation of International Atomic Time (Temps Atomique International; TAI) and UTC (Box 2). Both represent integrated timescales based on the time interval, the second, which is derived from the frequency of the ground-state hyperfine transition in ^{133}Cs. They are established by the International Bureau of Weights and Measures (BIPM) in Paris in a process where about 75 (mostly) national timing institutes contribute to this process with their clock ensembles. In a monthly bulletin (Circular T), the BIPM display the values of the differences between UTC and its local realizations at the given laboratory 'k', named UTC(k)[65]. Each week, the BIPM also publishes a rapid UTC (UTCr) approximation to UTC. OCNs provide the opportunity to deliver even faster information about the time offset with respect to UTC, which also would be of particular importance to the timescale of each global navigation satellite system.

Optical atomic clocks can contribute to TAI and UTC as secondary representations of the second[66]. (The updated recommendations can be found at http://www.bipm.org/en/publications/mises-en-pratique/standard-frequencies.html.) They can improve the stability of TAI and UTC. By replacing less accurate clocks such as Cs beam clocks, hydrogen masers, or commercial clocks in the generation of these timescales, optical atomic clocks will gradually begin to dominate these scales. It is likely that a new definition of the second based on an optical atomic clock transition will take place[21,67,68]. The first steps, using optical clocks for the generation of timescales, have already been taken[69–72]. To utilize the full power of the OCNs for timekeeping it will also be necessary to improve the models for determining the gravitational potential to the relevant uncertainty. Currently, differences in the gravitational potential can be modelled[55] to a few parts in 10^{18}, but in other more complex mountainous regions[73] only to a few parts in 10^{17}.

Delivering optical reference frequencies to any place connected by the fibre links[74] with a relative instability of 1 part in 10^{19} cannot be achieved by other methods based on microwave frequencies. Several experiments have made use of the opportunities provided by fibre links. A Sr lattice clock has been evaluated using a remote frequency standard[75]. In France, the link from LNE-SYRTE to the Laboratoire de Physique des Lasers has been employed to frequency stabilize a mid-infrared laser for laser spectroscopy in this region[76]. The French project REFIMEVE+ (www.refimeve.fr/) is currently setting up a national fibre network between 20 research laboratories (Fig. 4) to disseminate an ultrastable frequency signal for a wide range of applications[77]. Similarly, in Italy, the fibre infrastructure is being used for the delivery of a high-accuracy frequency reference to the European Laboratory for Non-linear Spectroscopy (LENS) for atomic physics experiments[78]. The links from PTB at Braunschweig to the Institute of Quantum Optics at Hannover and to the Max Planck Institute for Quantum Optics at Garching have been used to calibrate an optical Mg frequency standard[79] and the optical hydrogen 1S–2S transition[80], respectively.

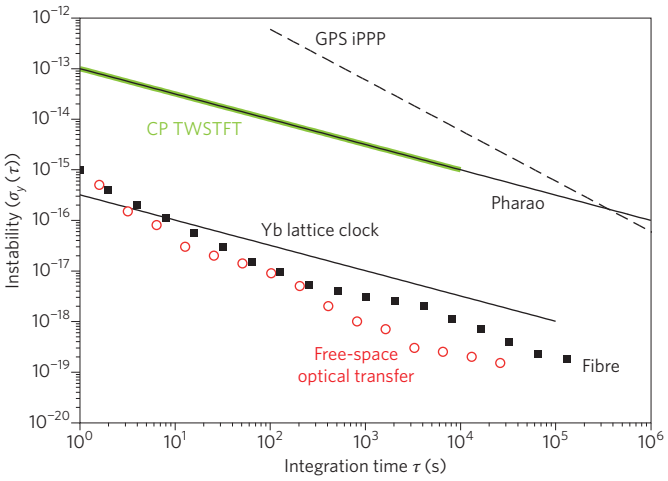


Figure 3 | Fractional instability of time and frequency transfer and clocks described by the Allan deviation (ADEV) or modified Allan deviation (mod ADEV) as function of the integration time τ. GPS iPPP, integer Precise Point Positioning technique[25] (mod ADEV); CP TWSTFT, fit to the carrier-phase two-way satellite time and frequency transfer data[29] (mod ADEV); Pharao, cesium clock for the Atomic Clock Ensemble in Space (ACES) mission on board the International Space Station[31] (ADEV); Yb lattice clock, fit to measured instability of an optical lattice clock[14] (ADEV); Fibre, fibre link between Paris and Braunschweig[55] (ADEV); Free-space optical transfer[40] (mod ADEV).

Relativistic geodesy is based on the fact that according to the theory of general relativity the frequency of a clock is affected by the local gravitational potential. When a clock is raised from the surface of the Earth by 1 m, the change in the gravitational potential results in a positive frequency shift of about 10^{-16}. This has been demonstrated

Box 2 | Frequency standards, clocks and timescales.

A microwave or optical oscillator whose frequency is controlled to correspond to the centre frequency of an unperturbed atomic transition provides a stable and continuous oscillating signal. If this frequency is known with respect to the frequency of a cesium clock that is used to realize the SI unit of hertz such a device is known as a frequency standard. If its frequency is converted to a sequence of pulses, for example, 1 pulse per second, the device represents an atomic clock. A timescale can be derived if a well-defined pulse is used as a starting point and the subsequent pulses are counted. Temps Atomique International (TAI; from which Coordinated Universal Time (UTC) is derived) is a continuous timescale generated by averaging over a large atomic clock ensemble. Its scale unit (the second as defined by the transition between the two hyperfine levels of the ground state of ^{133}Cs) is steered by the very best atomic ^{133}Cs clocks. Frequency comparisons between distant clocks can be achieved by the transfer of a single carrier frequency or by transportation of one of the clocks. Time transfer, on the other hand, will need a modulated signal with the information of the elapsed time or the transportation of a clock that is continuously operating during the transportation. Periodically repeated time transfers, of course, also provide frequency transfer, which is the standard method employed in the realization of TAI and UTC.

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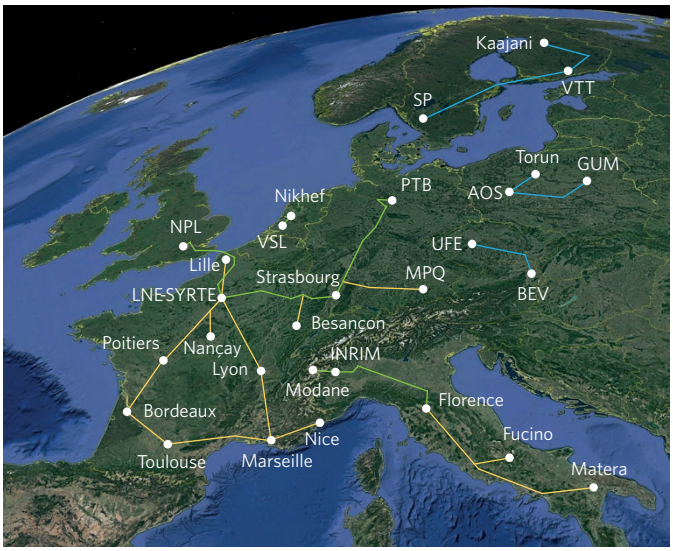


Figure 4 | Fibre links in Europe. Frequency links (green), frequency links commissioned (yellow) and time links (blue). AOS, Astrogeodynamic Observatory in Borowiec near Poznan, Poland; BEV, Bundesamt für Eich- und Vermessungswesen, Vienna, Austria; GUM, Główny Urząd Miar — Central Office of Measures, Warsaw, Poland; INRIM, Istituto Nazionale di Ricerca Metrologica, Turin, Italy; LNE-SYRTE, Laboratoire National de Métrologie et d'Essais — Système de Références Temps-Espace, Paris, France; MPQ, Max Planck Institute for Quantum Optics Garching/Munich, Germany; NPL, National Physical Laboratory, Teddington, UK; Nikhef, National Institute for Nuclear and High-Energy Physics, Amsterdam, the Netherlands; PTB, Physikalisch-Technische Bundesanstalt, Braunschweig, Germany; SP, Technical Research Institute of Sweden, Boras, Sweden; UFE, Institute of Photonics and Electronics of the Czech Academy of Science, Prague, Czech Republic; VSL, Dutch National Metrology Institute, Delft, the Netherlands; VTT, VTT-MIKES Metrology, Espoo, Finland. Image from Google Earth, US Dept of State Geographer, © 2016 Google, Image Landsat, Data SIO, NOAA, US Navy, NGA, GEBCO.

with two Al^+ optical atomic clocks in a laboratory experiment[81]. This so-called gravitational redshift[82] has to be corrected for any clock that contributes to TAI and UTC. With the smaller uncertainties of optical atomic clocks, however, this effect can be used the other way round to determine the Earth's gravitational potential more accurately. This emerging field requires (1) clocks that can probe the local gravitational potential with fractional accuracy approaching 10^{-18} and (2) links for their comparison that can support this accuracy, that is, OCNs. In the clock comparison experiment between Paris and Braunschweig[55], it was necessary to take into account a correction of 247.4 (4) $\times$ 10^{-17} resulting from the difference of the geopotential. The frequencies of three optical atomic clocks connected via a 30 km fibre with a height difference of 15 m have been compared and a centimetre-level chronometric levelling has been demonstrated[83].

Relativistic geodesy can support the measurements obtained by satellite gravimetry missions such as GRACE[84] and GOCE[85], and could support the measurements from the future GRACE Follow-on mission[86]. These missions provide quantifications for global and regional mass redistribution and a global coarse grid of absolute determinations of the gravitational potential. The GOCE satellite, for example, at its extremely low altitude, was able to determine the geoid with an uncertainty of 1–2 cm with a lateral resolution slightly better than 100 km. OCNs, on the other hand, could provide totally independent difference measurements of the gravitational potential to investigate the local small-scale variations with much higher resolution. OCNs can essentially help to define a more general geodetic fundamental network, including the accurate form and knowledge of the geoid, the height systems, the ocean surface, satellite orbits, or the tides. On the level of 10^{-18}, tidal effects will shift the frequencies of the local optical atomic clocks temporally with respect to remote optical atomic clocks. Future frequency comparisons between optical atomic clocks and real timescales based on a future definition with OCNs need to include improved modelling of these gravitational effects[83].

There are increasing applications for OCNs in space exploration and astronomy. Astronomical measurements with ultimate spatial resolution and positioning accuracy rely on radio astronomy with telescopes. Since the spatial resolution for a given wavelength of the radiation observed is proportional to the diameter of the used telescopes, often the telescopes are coupled in an interferometric arrangement to increase the resolution defined by the baseline between the telescopes. In these interferometers, the signals obtained in the single telescopes have to be correlated to allow for the subsequent evaluation. Previously, atomic clocks, mostly hydrogen masers, have been used for the necessary time stamps. The European Low Frequency Array (LOFAR; www.lofar.org) is a radio telescope where 50 stations with a large number of low-cost antennas are connected via the Internet into a single large telescope. It surveys the universe at frequencies from about 10–240 MHz. The Atacama Large Millimeter/submillimeter Array (ALMA; www.almaobservatory.org) has been in full operation since 2013 and comprises 66 high-precision antennas located at an altitude of 5,000 m in northern Chile. The antennas operating at frequencies between 30 and 950 GHz can be moved and positioned in distances from 150 m to 16 km and are connected with high-speed data links[87]. The Square Kilometre Array (SKA; www.skatelescope.org) is planned to be constructed between 2018 and the late 2020s and to have thousands of dishes with up to a million antennas at the sites in South Africa and Australia. Apart from applications in these large baseline interferometry arrays, ultrastable oscillators and links will be used in submillimetre-wavelength very large baseline interferometry (VLBI)[88]. The superior stability of optical atomic clocks can be delivered by OCNs. As an example, in Italy, frequency dissemination has been implemented by the Italian National Metrology Institute INRIM to the Medicina radiotelescopes, near Bologna, for VLBI antenna synchronization[46,89]. Better clocks or OCNs are needed in the future for the ground stations of NASA's (http://deepspace.jpl.nasa.gov) and the European Space Agency's[90] Deep Space Networks that support interplanetary spacecraft missions.

With regard to ultrasensitive measurements, it has been pointed out[91] that the availability of long-distance underground fibre-optic links opens a possibility of implementing interferometric fibre-optic gyroscopes enclosing very large areas. Such large-area fibre-optic gyroscopes on a multiplexed fibre network[92] represent interferometric sensors that could achieve high sensitivity to rotations and could be useful for seismic applications. OCNs may also

Box 3 | Uncertainty and instability.

Atomic clocks are characterized[105,106] by two different quantities. (1) The systematic 'uncertainty' of a clock is the estimated potential deviation of the clock frequency from the frequency ν_0 of the unperturbed reference transition (Fig. 1) after corrections for all known shifts. (2) The frequency 'instability' describes statistical fluctuations of the frequency delivered by a clock. It is investigated by comparing the clock's output signal with a more or similarly stable oscillator and calculating the so-called Allan deviation (ADEV) for various integration times τ of the data (Fig. 3). If the measured frequency instability is limited by white frequency noise, for example, by the quantum projection noise[107] of the atomic clock, the (modified) Allan deviation (mod ADEV) decreases with $\tau^{-1/2}$, while a limitation resulting from white phase noise is indicated by a ($\tau^{-3/2}$) τ^{-1} dependence.

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be used to monitor spatiotemporal changes in the geopotential caused by active volcanoes[83].

The search for 'new' physics beyond the standard model of particle physics such as for dark matter represents another possible application for OCNs. It has been proposed[93] to search for extended dark-matter objects with topological defects that might de-synchronize the initially synchronized clocks in an extended OCN when the dark-matter defect passes the OCN. When the dark matter forms an oscillating classical field it may also show up when a photon wavelength is compared with the length of an interferometer arm[94], for example, in a combination of a Sr optical lattice clock and a Si single-crystal cavity interferometer[95]. Instead of such a small-scale platform for these measurements, huge interferometers such as LIGO or VIRGO[96] could be used to increase the sensitivity. In such a case, OCNs would be needed if the best interferometers and the best optical atomic clocks are spatially separated. Many of the suggested experiments rely on an OCN similar to that employed in the Global Network of Optical Magnetometers for Exotic Physics (GNOME) project[97]. The GNOME project aims to synchronously measure the signals from optical magnetometers in distant locations to look for variations of the coupling between spins in the magnetometers and hypothetical fields. Such exotic fields and couplings — not predicted by the standard model — are postulated by a variety of theories. Even more far-reaching ideas on a quantum network of clocks[98] might build on established OCNs. It has been pointed out[99] how the precision of connected remote optical atomic clocks can be enhanced. Phase-coherently-linked optical atomic clocks can also be used to reduce the clock laser noise in proposed applications to detect gravitational waves with atom interferometers[100].

Conclusion and outlook

A new era of OCNs has begun with several examples realized in Japan, the US, different European countries and elsewhere. An estimated number of 40 optical atomic clocks all over the world that outperform the best cesium clocks in terms of accuracy and stability, and several more under development, asks for better linkage worldwide. Fibre links connecting England, France and Germany, or France and Italy, or between different sites in Japan have been established. OCNs have been used for comparisons with a performance far beyond that of the established satellite-based time and frequency methods and have largely contributed to new knowledge and opened novel technological opportunities. Nevertheless, the conventional technologies can also be improved, for example by using larger bandwidth. Free-space optical links or higher-quality clocks on board satellites promise further progress but are still at an experimental stage.

The best optical atomic clocks are beginning to move from the prototype stage. Up to now, operation longer than a few days was not necessary, but with the OCNs available, the long-term operation will improve, which is also a prerequisite for disseminating stable frequencies on a regular basis offering also commercial services. No technological limitations for long-term operation can be envisaged, for example, a local fibre network at the site of PTB continuously delivers an extremely stable and reliable near-infrared reference frequency for different onsite buildings. There is, however, a trade-off between further continuous improvement of the best optical atomic clocks and freezing their technology at a given point to end up with a well-engineered optical atomic clock with higher technology readiness. Several attempts are underway to set up such optical atomic clocks. At present, the few existing fibre links in an OCN need to be maintained by humans. Furthermore, careful characterization and monitoring of the noise cancellation and stabilization components is required. There are, however, commercial components for the dark-fibre and the dark-channel approaches available.

In the near future, there will be several regional OCNs such as those shown in Fig. 4. Even though conventional fibre networks are not yet ready for two-way phase-coherent amplifier stages, they can be refurbished easily since the active components of existing networks are replaced to allow for installation of the next-generation components. Thus it can be foreseen that the regional OCNs will grow within the coming years.

In the long term, a global OCN linking the regional OCNs with the different methods outlined above that work in a hybrid arrangement can be envisaged. Very good clocks on board satellites and optical free-space links as well as improved microwave links have the potential to link the continental OCNs on a transcontinental scale. In the short term, however, frequency comparisons over continents with fractional inaccuracies near or below 10^{-17} will probably have to rely on only two methods: transportable optical clocks and optical frequency ratios. With about a handful of transportable optical atomic clocks already ready or under construction, continental OCNs can be linked. Purely optical frequency ratios of optical atomic clocks can be measured with femtosecond laser frequency combs in principle with an unprecedented low fractional uncertainty of 10^{-20} and below[101]. Several direct optical frequency ratios between the different optical frequency standards have been determined within the 10^{-17} range for Al^+/Hg^+ (ref. 102), Hg/Sr (ref. 103), or Yb/Sr (ref. 104). It might also be possible to compare the measured optical frequency ratios of two spatially separated OCNs with the same clock ensemble thereby establishing a 'virtual' link even if they are not directly linked.

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Competing financial interests

The author declares no competing financial interests.

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